

D800

Human-Computer Interaction

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

After a user taps Upload on a 200 MB video, the screen shows nothing for 40 seconds, and many users tap Upload again, creating duplicate posts. The app was built with a cross-platform framework and has a 4.6-star rating. Which usability heuristic is most directly violated?

- A. Consistency and standards
- B. Rebuild in native code, since cross-platform frameworks cause the silence.
- C. Aesthetic and minimalist design
- D. Visibility of system status

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

In a banking app, the Transfer button is used in 80% of sessions but is a 24-pixel icon in the top corner, far from the thumb. The brand color is teal, and the app was rebuilt last year. Which change most reduces the time it takes to tap Transfer?

- A. Recolor the button in a brighter shade of teal.
- B. Shrink it further to reduce clutter at the top.
- C. Make the button larger and move it into the thumb's easy-reach zone near the bottom.
- D. Add a confirmation screen before every transfer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Early in a project for a library system's room-booking site, the team has three competing page layouts. The library has 12 branches and a brand guide with exact colors and fonts. What should the team produce first so users can compare the layouts?

- A. Pixel-perfect mockups in the brand colors and fonts.
- B. A fully coded version of each layout.
- C. Low-fidelity wireframes of each layout showing structure and flow.
- D. Skip wireframes, because low-fidelity work is never useful once a brand guide exists.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An eye-tracking study of a checkout page shows a hot spot on the promo-code field: long fixations and repeated returns. The study used a 60 Hz tracker and 12 participants in a lab with blue walls. What is the best interpretation?

- A. The field draws attention and may be causing confusion or distraction; follow up with think-aloud or interviews to learn why.
- B. Users like the promo field, so it should be made bigger.
- C. It proves users understood the field.
- D. The results are invalid because of the lab's wall color.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A clinic's patient portal marks required form fields only by turning their labels red. The portal serves 30,000 patients and was redesigned in a two-week sprint. Which change best fixes the accessibility problem?

- A. Add a visible text cue such as "Required" and mark the field as required in the code so screen readers announce it.
- B. Show a tooltip explaining the red labels when the mouse hovers.
- C. Use a brighter, more saturated red.
- D. Redo the redesign with a longer sprint.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Gray body text on a white background has a contrast ratio of 3.2:1. What does WCAG 2 Level AA require for normal-size text?

- A. At least 4.5:1, so this text fails and must be darkened.
- B. At least 3:1, so this text passes.
- C. At least 7:1, so this text fails.
- D. It passes, since any ratio above 1:1 is readable.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Halfway through a usability session run by a university lab on a state grant, a participant says she is uncomfortable and wants to stop. She signed the consent form and was promised a \$25 gift card. What should the researcher do?

- A. Remind her that she signed the form and ask her to finish.
- B. Stop the session right away and give the promised compensation as the approved protocol states.
- C. Stop the session but keep the gift card, since she did not finish.
- D. Check with the grant funder before deciding.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A study stores participants' names and emails in a locked file linked to ID codes, and reports results only by ID code. How should this data be described?

- A. Public, since results are reported
- B. Exempt from all consent, because ID codes are used
- C. Anonymous
- D. Confidential

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Seven participants completed a checkout task in 42, 45, 48, 51, 55, 58, and 240 seconds; the last participant took a phone call mid-task. The sessions were recorded on a 13-inch laptop, and all participants came from a customer panel. Which value best represents typical time on task?

- A. 77 seconds (the mean)
- B. 240 seconds
- C. 198 seconds (the range)
- D. 51 seconds (the median)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Eight participants attempted a task, and the report lists a 62.5% completion rate. How many participants completed it?

- A.** 3
- B.** 8, since any rate over 50% means everyone finished.
- C.** 5
- D.** About 13

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A participant, a 22-year-old design student, completed the System Usability Scale after a 40-minute session. Responses to items 1, 3, 5, 7, 9 were 4, 5, 4, 4, 5; responses to items 2, 4, 6, 8, 10 were 2, 1, 2, 1, 2. What is the SUS score?

- A.** 34
- B.** 85
- C.** 15
- D.** 30

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A participant's SUS score is 70. What was the sum of the adjusted item contributions?

- A.** 175
- B.** 28
- C.** 68
- D.** 70

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

After a user taps Upload on a 200 MB video, the screen shows nothing for 40 seconds, and many users tap Upload again, creating duplicate posts. ~~The app was built with a cross-platform framework and has a 4.6 star rating.~~ Which usability heuristic is most directly violated?

Target word: which heuristic **Decides:** No feedback for 40 seconds after an action.

A	Confusion set	About matching conventions; the problem here is missing feedback.
B	Camouflage	The framework does not decide the missing feedback; any stack can show progress.
C	Adjacent category	About clutter; adding a progress bar is the fix, not removing elements.
D	Correct	The system gives no sign it is working, so users retry.

2. Answer C Concept 2

In a banking app, the Transfer button is used in 80% of sessions but is a 24-pixel icon in the top corner, far from the thumb. ~~The brand color is teal, and the app was rebuilt last year.~~ Which change most reduces the time it takes to tap Transfer?

Target word: change that speeds acquiring the target **Decides:** Frequent target that is small and far away.

A	Camouflage	Color may aid noticing but does not change size or distance.
B	Inverted operation	Smaller targets take longer to hit.
C	Correct	Bigger and closer targets are faster to hit.
D	Adjacent category	Error prevention adds a step; it does not make the button faster to reach.

3. Answer C Concept 3

Early in a project for a library system's room-booking site, the team has three competing page layouts. ~~The library has 12 branches and a brand guide with exact colors and fonts.~~ What should the team produce first so users can compare the layouts?

Target word: right artifact for early layout comparison **Decides:** Early stage with competing structures.

A	Camouflage	The brand guide does not decide the stage; polish invites comments on looks.
B	Adjacent category	A working build is costly and premature for comparing three structures.
C	Correct	Cheap, fast, and keeps feedback on structure rather than looks.
D	Absolute language	A brand guide does not settle layout; wireframes still test structure.

4. Answer A Concept 4

An eye-tracking study of a checkout page shows a hot spot on the promo-code field: long fixations and repeated returns. The study used ~~a 60-Hz tracker~~ and 12 participants ~~in a lab with blue walls~~. What is the best interpretation?

Target word: what the eye data supports **Decides:** Eye tracking shows attention, not reasons or understanding.

A	Correct	Fixations show where people look; other methods explain why.
B	Confusion set	Attention is not preference; long looks often signal difficulty.
C	Absolute language	Looking does not prove comprehension.
D	Camouflage	Wall color does not decide the interpretation.

5. Answer A Concept 5

A clinic's patient portal marks required form fields only by turning their labels red. ~~The portal serves 30,000 patients and was redesigned in a two-week sprint.~~ Which change best fixes the accessibility problem?

Target word: fix for a color-only cue **Decides:** Meaning is carried by color alone.

A	Correct	Removes reliance on color and makes the state perceivable to all users.
B	Adjacent category	Hover is not available to keyboard, touch, or screen-reader users.
C	Dropped element	Still color alone; people with color-vision differences or screen readers still miss it.
D	Camouflage	Sprint length does not decide the fix.

6. Answer A Concept 5 • transfer

Gray body text on a white background has a contrast ratio of 3.2:1. What does WCAG 2 Level AA require for normal-size text?

Target word: reverse: from a measured ratio to the pass/fail rule **Decides:** AA for normal text is 4.5:1.

A	Correct	3.2:1 is below the 4.5:1 AA threshold for normal text.
B	Confusion set	3:1 applies to large text and UI components, not body text.
C	Adjacent category	7:1 is the stricter AAA level.
D	Absolute language	AA sets a specific minimum.

7. Answer B Concept 6

Halfway through a usability session ~~run by a university lab on a state grant,~~ a participant says she is uncomfortable and wants to stop. She signed the consent form and was promised a \$25 gift card. What should the researcher do?

Target word: response to a withdrawal request **Decides:** Consent is ongoing and withdrawal carries no penalty.

A	Inverted operation	Treats consent as a one-time lock instead of an ongoing choice.
B	Correct	Participants may withdraw at any time without penalty.
C	Dropped element	Drops the "without penalty" part of withdrawal.
D	Camouflage	The funder does not decide a participant's right to stop.

8. Answer D Concept 6 • transfer

A study stores participants' names and emails in a locked file linked to ID codes, and reports results only by ID code. How should this data be described?

Target word: reverse: from the handling to the label **Decides:** Identities are known to the researchers but protected.

A	Adjacent category	Aggregate reporting does not make individual data public.
B	Absolute language	Coding data does not remove consent requirements.
C	Confusion set	Anonymous means no one, including researchers, can link data to a person.
D	Correct	The team can link data to people but keeps that link private.

9. Answer D Concept 7

Seven participants completed a checkout task in 42, 45, 48, 51, 55, 58, and 240 seconds; the last participant took a phone call mid-task. ~~The sessions were recorded on a 13-inch laptop, and all participants came from a customer panel.~~ Which value best represents typical time on task?

Target word: typical time on task **Decides:** One outlier skews the mean.

A	Confusion set	$539 \div 7 = 77$ is inflated by the one outlier.
B	Wrong stem number	The outlier itself.
C	Adjacent category	Range measures spread, not a typical value.
D	Correct	The middle of seven sorted times; not pulled up by the 240-second outlier.

10. Answer C Concept 7 · transfer

Eight participants attempted a task, and the report lists a 62.5% completion rate. How many participants completed it?

Target word: solve for completions **Decides:** Completions = 0.625×8 .

A	Adjacent category	That is the number who did not complete.
B	Absolute language	A rate is a share, not a pass/fail for the group.
C	Correct	$0.625 \times 8 = 5$.
D	Inverted operation	$8 \div 0.625$ divides instead of multiplying.

11. Answer B Concept 8

A participant, ~~a 22-year-old design student,~~ completed the System Usability Scale ~~after a 40-minute session.~~ Responses to items 1, 3, 5, 7, 9 were 4, 5, 4, 4, 5; responses to items 2, 4, 6, 8, 10 were 2, 1, 2, 1, 2. What is the SUS score?

Target word: SUS score **Decides:** Odd: $3+4+3+3+4 = 17$; even: $3+4+3+4+3 = 17$; total 34.

A	Dropped element	Forgets the $\times 2.5$ step.
B	Correct	$34 \times 2.5 = 85$.
C	Inverted operation	Applies the even-item rule to odd items and vice versa: 6×2.5 .
D	Wrong stem number	The raw sum of responses with no item adjustment.

12. Answer B Concept 8 · transfer

A participant's SUS score is 70. What was the sum of the adjusted item contributions?

Target word: solve backward for the adjusted sum **Decides:** $SUS = \text{sum} \times 2.5$.

A	Inverted operation	Multiplies by 2.5 instead of dividing.
B	Correct	$70 \div 2.5 = 28$.
C	Confusion set	The often-cited average SUS score, not this participant's sum.
D	Dropped element	Ignores the scaling step.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Usability heuristics	Standard heuristics name recurring design failures; silence after an action violates visibility of system status.	Naming any heuristic that sounds related instead of the one the symptom points to.	Users repeat an action or wait with no feedback.
2	Target size and distance (Fitts's law)	Time to hit a target rises with distance and falls with target size, so frequent actions should be big and close.	Shrinking or recoloring a control to make it faster.	A frequently used control that is small or far from the pointer or thumb.
3	Wireframe fidelity	Use low-fidelity wireframes early to compare structure and flow; add visual polish only after the structure is settled.	Polishing colors and fonts before layouts are tested.	"Early," "compare layouts," or "several competing options."
4	What eye tracking shows	Eye tracking shows where and how long people look (fixations), not why or whether they understood; pair it with think-aloud or interviews.	Reading a heat map as proof of liking or comprehension.	Long fixations or repeated returns to one element.
5	Accessibility (POUR and contrast)	Content must be perceivable, operable, understandable, and robust: never convey meaning by color alone, and normal text needs at least 4.5:1 contrast at WCAG AA.	Fixing a color-only cue with a different color, or using the 3:1 large-text threshold for body text.	Meaning carried only by color, or a contrast ratio in the stem.
6	Informed consent with human subjects	Consent is voluntary and ongoing: participants may withdraw at any time without penalty, and identifiable data kept private is confidential, not anonymous.	Treating a signed form as locking the participant in, or calling coded data anonymous.	A participant wants to stop, or names are stored with a key.
7	Summarizing test results	Task success rate = completions ÷ attempts; use the median for time on task when times are skewed by outliers.	Reporting a mean pulled up by one extreme time.	One task time far larger than the rest.
8	SUS scoring	System Usability Scale: odd items score (response – 1), even items score (5 – response); sum and multiply by 2.5 for a 0–100 score.	Summing raw responses or forgetting the 2.5 multiplier.	Ten 1–5 ratings and a request for a SUS score.

Course scope (public description): Human-Computer Interaction is a course that covers the principles of human-computer interaction (HCI), incorporating insights from psychology, design, and engineering to understand user interactions. This course prepares students to design and evaluate user-centered systems using tools like wireframing and eye tracking. The course addresses ethical implications, focusing on privacy, consent, and accessibility. Additionally, the course helps students develop basic data analysis skills for interpreting user testing results and complete the CITI Institutional Review Board (IRB) Protocol Review for working with human subjects. The main goal of the course is to teach students how to create intuitive, accessible, and engaging interactive systems informed by user testing.